Kalathur Chenchu Kishore Kumar

Computer Vision & ML Engineer, Valeo R&D

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Education

M.Tech in Artificial Intelligence **July 2021 - Jun 2023**
Indian Institute of Technology, Jodhpur *CGPA: 7.56/10*

B.Tech in Computer Science and Engineering **July 2015 - May 2019**
Sree Vidyanikethan Engineering College, Andhra Pradesh *CGPA: 8.02/10*

Professional Experience

Perception Engineer, Valeo, Chennai **February 2026 - Present**

- Developing **camera-based perception modules** for **ADAS and ARAS systems** using deep learning, enabling robust **object detection and scene understanding** in real-world driving environments for automotive safety applications.

Gen AI Developer, Senior Engineer, Wipro, Bengaluru **August 2025 - February 2026**

- Leading development of **Generative AI applications**, integrating **LLMs**, **Vision-Language Models**, and **Diffusion Models** for enterprise automation and multimodal data understanding.
- **Vision-Language Models and Medical Imaging** - Fine-tuned **VLMs** to improve visual detection and structured reasoning, and developed deep learning based responsi models for medical image analysis.

Computer Vision & ML Engineer, Technical Lead, HCLTech, Chennai **October 2023 - August 2025**

- Built a **3D scene reconstruction pipeline** with **NeRF** and **COLMAP** from **monocular camera** frames, enhancing environment modeling and enabling precise in-house equipment installation.
- Explored and implemented **Stable Diffusion Model** to generate images from **facial landmarks** and vice-versa while incorporating **ControlNets** for precise control over prompt text and depth maps from landmarks to generate image.
- Developed a comprehensive **Alert-Monitoring system** to track unsafe driving behaviors, optimizing & **quantizing** the model for **edge devices** with **OpenVINO**, achieving an **83% reduction in model size** & **60 FPS** performance.
- Developed a method for **detecting periodic patterns** in semiconductor images using **frequency domain techniques**, **wavelet decomposition**, and **histogram equalization**, achieving a **proof of concept** with an **IoU score of 80%**.
- Mitigated **catastrophic forgetting** in **incremental object detection** via **knowledge distillation**, achieving 85% **mAP** on unseen **COCO classes**, comparable to standard object detection models.

Publications

Curriculum-Guided Massive Multi-Agent System for Long-Horizon Reasoning (Paper Link) **2025**

- Proposed a hierarchical **64×64 grid multi-agent architecture** to decompose long-horizon reasoning tasks while reducing computational cost.
- Introduced **spatial curriculum learning** with **Negative Log-Likelihood (NLL) confidence calibration** and a **Thompson Sampling-based curriculum manager** for adaptive training.
- Demonstrated improved stability and reduced oracle dependency on the **Tower of Hanoi long-horizon reasoning benchmark**.

Technical Skills

Programming: Python, C/C++, Java.
Machine Learning Frameworks: Scikit Learn, PyTorch, OpenCV, LangChain, LLamaIndex, RAG, LORA & QLORA.
Technical: Machine Learning, Deep Learning, Video & Image Processing, Computer Vision, NLP, GenerativeAI, LLM.

Relevant Coursework

- Advanced Deep Learning
- Artificial Intelligence
- Blockchain & Security
- Computer Vision
- Computer Graphics
- Cognitive Neuroscience
- Machine Learning
- ML-Ops & DL-Ops
- Non-linear programming
- Financial Engineering
- Virtualization & cloud computing

Academics Accomplishments & Extracurricular

- Presented a **Mixed Reality** concept at **INAE-SERB Youth Conclave 2022**, IIT Jodhpur.
- Attended workshops on **Ethical Hacking & Information Security** (IISC Bangalore, Jan 2018) and **Google Android Development** (IIT Bombay, 2017-2018).
- 96.68^{th} Percentile in **GATE 2021**.
- TA for courses: Deep Learning, DLOps, Security and its Applications.
- Managing **Student Wellbeing Committee** web portal and providing **student guidance** at **IIT Jodhpur** (2022).